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Unit Name: Advanced Artificial Intelligence Research Topics COMP6011

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REPORT 2: TOPIC 2

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Abstract—This report investigates the use of Large Language models (LLMs) for suicide risk detection from conversational data. The proposed system is made as an early screen tool and for triage support tool as well. The system identifies linguistic and behavioral risk signals and refers high risk scenarios to qualified healthcare professionals. A literature review and benchmarking is done across suicide risk classification, crisis text counseling, mental health LLMs, and evidence extraction benchmarks. The findings show that LLM show promising results for contextual reasoning and explanation. Simple prompt-based system was found to be limited due to hallucination, prompt sensitivity, and possible false negatives. A proof of concept was implemented using a local LLM deployed through Ollama on an edge device. The report recommends a privacy enabled hybrid LLM architecture using local inference. Further validation on larger, diverse, clinically reviewed datasets is required before real-world deployment.

Index Terms—Computer Society, IEEE, IEEEtran, journal, L^AT_EX, paper, template.

1 INTRODUCTION

MENTAL health disorders are one of the most important problems of the twenty first century [1]. More than one billion people are now estimated to be living with mental health problems globally [1]. Mental health problems like depression, anxiety, trauma affect hundreds of millions of people all over the world [1]. Suicide also remains as one of the most preventable cause of death with more than 720,000 deaths each year [2]. Suicide is also ranked as third leading cause of deaths among people aged between fifteen and twenty nine [2]. Early detection of suicide risk is an important objective in healthcare system because suicide is prevented through early warning signs before fatal escalation [2], [3]. However, immediate access to mental health professionals remain limited due to workforce shortages, stigma, financial barriers and increasing service demand [4], [5]. These problems creates delays in immediate assessment and intervention of mental health problems.

Recent developments in Large Language Models (LLMs) and conversational artificial intelligence have created new opportunities for mental health assessment, support and triage [6], [7], [8]. Modern LLMs are able to understand different languages, complex emotional behavior, context and subtle changes on an ongoing conversation [9]. This features has made LLMs extremely important and relevant to mental health education, assessment, and intervention [6], [9]. In comparison to older rule based system or keyword detectors for suicide based words, LLMs can work with large variation of languages and conversational context [8]. These features are important because suicide risk may be shown with indicators like burden, social withdrawal, lack of future, indirect self harm references, and worsening emotional tone [10]. These features also make LLMs suitable candidates for early detection of suicide risk.

However, the application of LLMs in high risk mental health situations introduces technical, ethical, and gover-

nance challenges [6], [7]. Misclassifications can produce harmful false negatives. The system can fail to identify a person at risk or low-risk people are incorrectly escalated for intervention [8]. Additional problems would include hallucinated outputs, demographic bias, privacy protection, informed consent and compliance [11]. These factors are important when considering implementing AI in healthcare systems [11].

The aim of this research paper is to investigate how LLM based systems can be used as an early screening system for suicide risk detection from conversational transcripts.

2 LITERATURE REVIEW

2.1 Rule-Based and Keyword Detection Systems

Rule based systems for early detection of suicide is one of the simplest form of computational risk detection. These systems search for predefined words, phrases or patterns related to suicide risk [10], [12]. This can include references to death, self-harm, methods, intent, time, hopelessness or being a burden [10], [12]. This system has a simple architecture where the system will receive a text input. It will then apply a dictionary or rule set, count and weight matched risk indicators and produce a risk flag.

A major advantage in this system is in its explainability. If a system flags a user for high suicidal risk, the reason for it can be looked back on to exact matched words or phrases. Explainability is useful in healthcare because health practitioners can understand and see why the alert was triggered. Rule-based and lexicon based systems are also quite inexpensive and computationally efficient [12]. These models can be used as a baseline in benchmarking to compare other complex models [12].

The disadvantage of this system can be seen in its poor contextual understanding. The system just uses keyword matches, so it will miss phrases like "I just want to disappear" or "everyone would be better without me". It can also falsely flag safe texts that discusses suicide in an academic, satirical, or fictional way. This limitation is supported by studies showing keyword matching for suicidal thoughts

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generating large amount of false alarm for distinguishing risk and non risk [13].

2.2 Classical Machine Learning Models

Classical Machine learning models have improved from Rule-Based lexicon systems by learning statistical patterns from labeled texts. Common models of these systems include Support Vector Machines (SVM), Logistic Regression, Naive Bayes, Random Forests, and ensemble models [14], [15], [16]. These models have been used extensively in early suicide risk detection [14], [15], [16]. These systems utilize feature extraction methods like F-IDF, n-grams, sentiment scores, topic features, or psycholinguistic features such as LIWC categories [14], [17]. The architecture of this system is usually text preprocessing, feature extraction, classifier training, and risk prediction. For instance, a text conversation will be transformed into a TF-IDF vector, then passed into an SVM or Logistic Regression classifier that predicts a suicide risk label. This type of logic or pipeline is very common in social media suicide detection research. This because social media conversations are in short texts, and this text can be converted into sparse lexical or statistical vectors for supervised classification [18]. For example, a report showed a macro-F1 of around 0.46 for an SVM baseline on Reddit suicide risk assessment, while a CNN model achieved around 0.42 [19]. This shows that classical models can sometimes outperform more complex neural models on limited datasets [19].

These models also have disadvantages in terms of understanding deep contextual meaning. It treats most conversations as statistical features, so it may not understand conversational context, implied meaning, sarcasm, or changes in emotional tone across a dialogue. The feature engineering must be done manually and this can fail in different languages [14], [15], [16]. For instance, research papers showed that feature engineered machine learning models could support user level suicide risk classification, but the task remained challenging because risk labels were difficult to separate properly [19].

2.3 Deep Learning Models

Deep learning based models were introduced to capture more complex language patterns than classical machine learning. Models utilising deep learning include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), Gated Recurrent Units (GRUs), and attention-based models [20]. These models have been applied in suicide risk assessment and mental-health text classification research [20]. In CNN based text classifiers, the model will detect important local phrase patterns using convolution filters over word embeddings. The text classifier model can identify short risky phrases, like those involving suicide threats and self-harm [21]. But CNN models have limited capability to understand long relationships in the entire dialogue [21]. RNNs, LSTMs, and GRUs work well with sequential data because they process text in order. This is helpful when analyzing risk in terms of suicide because there might be a gradual increase in risk through the course of interaction. A person might start by expressing general

suffering, then progress to hopelessness, and then talk about how they want to fade out or have no future left to look forward to. Sequence model can work with this progression better than other CNN models as these models have been widely explored for suicidal risk detection from social media text [21], [22]. Other models have made use of attention based CNNs. In this architecture, the model starts by encoding words into sentences, then encoding sentences into a user conversation level representation [23], [24]. The use of attention scores is helpful in identifying which part of the conversation led to the final result [23], [24]. This makes them more suitable for cases that require suicide detection at a conversation level instead of a sentence level [23], [24].

2.4 Transformer Models

Transformer based encoder models like BERT and RoBERTa are a breakthrough in NLP classification tasks. These transformer models uses self-attention to capture the connections between the tokens present within the text. In contrast to LSTMs, which only process the text sequentially, transformers can focus on multiple parts of the input at once, allowing them to learn bidirectional contextual representations [24], [25]. BERT is a model that was developed specifically for the task of pre-training deep bidirectional representations based on both left and right contexts [24], [25]. while RoBERTa further refined the BERT-like pre-training strategy through various optimizations [24], [25]. This is quite useful for predicting suicide risk. Although the phrase “I’m done” by itself does not necessarily indicate danger, the presence of additional context such as hopelessness, loneliness, and talk of dying changes the risk assessment. The transformer encoder, as opposed to a keyword based system is effective because it can generate contextualized representations of words based on surrounding text rather than treating terms as isolated statistical features [24], [25].

This architecture involves taking the input text and performing tokenization on it. It then feeds this through a pre-trained transformer encoder which transforms it into a contextualized representation. It then sends it to a classification head. The model is fine-tuned using labeled suicidal risk data. This type of architecture is common in suicide risk and crisis text classification involving transformers due to its ability to adapt pre-existing linguistic knowledge to the domain [26], [27].

For the text-based crisis encounters gathered using the SafeUT crisis counselling system, a report utilized a neural model based on the RoBERTa architecture [26]. The performance of their neural model included a ROC AUC score of 90.37, a sensitivity of 62.02, and a specificity of 92.89, whereas the TF-IDF Logistic Regression baseline resulted in a slightly smaller AUC score of 88.18 and a considerably smaller sensitivity of 39.52 [26]. It can be observed that a trade-off exists in clinical settings where the transformer-based neural models are able to predict more true positive instances, but can also cause higher false positives [26]. Transformer based encoders are good options for the categorization of suicide risk due to their better control compared to generative LLMs. It also has higher success in classification tasks. However, they would need labeled data and might find it difficult to handle long dialogues beyond

the number of tokens supported by the model [26]. In the context of high-risk mental health applications, such models need to be assessed not only through AUC and accuracy metrics but also sensitivity, specificity, false negatives, and error calibration [26].

2.5 Prompt Based Large Language Model

Prompt based LLMs are large pre-trained models which includes the GPT-4 family, Claude class, Gemini class, LLaMA, Mistral, and Mixtral. These models have zero training for specific tasks but utilizes prompts with data to produce output. In place of training a dedicated classifier, the model receives prompts that instruct the model on how to define the risk categories and perform the classification [7]. In recent studies on screening for suicide risk, researchers have used clinical and clinical inspired risk frameworks like the Columbia-Suicide Severity Rating Scale and Nurses' Global Assessment of Suicide Risk in evaluating prompt-based approaches [28], [29]. The prompt can also ask the model to identify risk indicators such as hopelessness, burdensomeness, intent, preparation, emotional distress, social support, or protective factors. For instance, the CLPsych 2024 Xin-Hai prompting system instructed the model to detect risk signals, classify the risk level, and produce outputs with structured data fields, such as the evidence extracted, label, and summary [30].

Prompt-based LLMs offer various benefits. These include quick prototyping capabilities, zero need for a task-specific training data set, and the ability to understand multiple languages. They are also versatile capable of classification, explanation, summarization, and generating structured JSON output. Literature reviews indicate that LLMs have potential in early screening, digital interventions, conversational agents, and mental health text analysis [7].

However, prompt-based LLMs have serious limitations in clinical contexts. There is potential for variability in the output from the LLMs based on the wording of the prompts, the temperature, type of prompting, and the version of the LLM. The German NGASR crisis-line study demonstrated variability in terms of reliability and consistency between human responders and the LLM based on zero-shot, few-shot, chain-of-thought, and temperature scenarios, with the reliability declining at higher temperature levels and several of the moderate risk categories reaching near chance levels of accuracy [29]. LLMs can also hallucinate explanations, overstate certainty, or produce unsupported reasoning, and cloud-based API use raises privacy and data-governance concerns in sensitive mental-health settings [7].

3 BENCHMARKING RESULTS

The objective of benchmarking is to understand how current AI and LLM models perform suicide risk and mental health prediction before developing the proof-of-concept application. Benchmarking data presented in this section is based on various studies conducted for suicide risk prediction. Benchmarked approaches include traditional machine learning, deep learning, transformer encoders, prompting LLMs, instruction-tuned LLMs, and hybrid LLMs.

3.1 Datasets

3.1.1 UMD / CLPsych Reddit Suicide Risk Dataset

The UMD Reddit Suicidality Dataset and CLPsych shared tasks are two of the most widely recognized publicly available benchmark datasets for predicting suicide risk based on online text data [19]. The UMD dataset was developed for identifying suicide risk based on online Reddit posts, and was intended to allow both human and machine evaluations of suicide risk at the user level [19]. This dataset was formed from Reddit users who posted in suicide related communities, and was based on a suicidal rating scale [19].

There were 11,129 possible users to annotate from the r/SuicideWatch dataset, along with 1,556,194 posts [19]. However, a sample of users from the dataset were annotated using structured annotation. From the dataset, there were 934 users annotated, 245 of whom had annotations provided by both expert and crowd annotators [19]. The results of the annotation revealed that there is significantly higher agreement among experts compared to crowd members [19]. Krippendorff's alpha equal to 0.81 for experts and 0.55 for crowdworkers [19]. This benchmark is very relevant since it focuses on the risk assessment of the user and not the individual sentences used to identify their sentiment level. In simpler terms, the model has to factor in the history of posts made by the user and not just analyze individual sentences. However, the data remains publicly sourced from social media and not from the private clinical interviews that take place or from the human-AI conversations.

3.1.2 SafeUT Crisis Counselling Dataset

SafeUT crisis counseling data is very significant since it makes use of a text-based crisis counseling session, and not posts on social media sites that have been made publicly available [26]. It analyzed 5,992 de-identified text-based crisis counseling sessions consisting of 273,804 messages collected using SafeUT [26].

3.2 Benchmarking Candidates

Table 1 summarises the candidate model families considered for benchmarking. The evaluation metrics for this benchmarking would include, Accuracy, Precision, Sensitivity, Specificity, Macro-F1, ROC AUC.

3.3 Quantitative Benchmarking

The benchmarking results summarised in Table 2 are based on prior suicide-risk detection studies, including CLPsych 2019, SafeUT crisis counselling, CLPsych 2024, privacy-preserving clinical LLM research [26], [31], [32], [33].

3.4 Qualitative Analysis

The results indicate that it is considerably more difficult to detect suicide risk than to perform basic sentiment analysis. As seen from the CLPsych 2019 results on Reddit, binary screening outperforms multi-class analysis. In CLPsych 2019 system paper, the best four-class model obtained a macro-F1 score of 0.379, while the best binary flagged-risk model obtained F1 = 0.836 and F1 = 0.736 for urgent-risk screening [31]. This result is understandable because categorizing

TABLE 1
Benchmarking Candidate Models for Suicide Risk Detection

Candidate	Model Type	Reason for Inclusion
TF-IDF + Logistic Regression / SVM	Classical ML	Strong transparent baseline for text classification.
Naive Bayes / Ensemble ML	Classical ML	Common in earlier Reddit suicide-risk benchmarks.
CNN / LSTM / Attention Models	Deep learning	Captures phrase patterns and sequential context.
BERT / RoBERTa	Transformer encoder	Strong classification models for mental-health text.
Prompt-Based LLMs	Decoder-based LLM	Useful for zero-shot/few-shot classification and explanation.

TABLE 2
Benchmarking Results from Suicide-Risk Detection Studies

Benchmark	Dataset / Task	Model	Results	Interpretation
CLPsych 2019 Reddit	Reddit user-level risk classification	Ensemble ML	Macro-F1 = 0.379	Suicide-risk classification into four levels is challenging. [31]
CLPsych 2019 Reddit	Binary flagged-risk screening	Naive Bayes	F1 = 0.836	It is simpler to detect binary risks than multi-level risks. [31]
CLPsych 2019 Reddit	Binary urgent-risk screening	Naive Bayes	F1 = 0.736	The detection of high-risk instances is possible yet not accurate enough. [31]
SafeUT crisis counselling	5,992 crisis-text encounters	RoBERTa Neural model	AUC = 90.37; sensitivity = 62.02; specificity = 92.89; precision = 59.52	Neural models improve high-risk sensitivity [26]
SafeUT crisis counselling	5,992 crisis-text encounters	TF-IDF Logistic Regression	AUC = 88.18; sensitivity = 39.52; specificity = 97.84; precision = 75.56	Classical approach has high accuracy but fails to detect many high-risk instances. [26]
CLPsych 2024	Reddit suicide-risk evidence extraction	LLM with knowledge self-generation	Recall = 93.5%; harmonic mean = 92.3%; consistency = 96.1%	LLMs have the potential for high performance when constrained. [32]
Privacy-preserving clinical text	Psychiatric admission notes	Locally hosted German Llama-2	Accuracy = 87.5%; sensitivity = 83.0%; specificity = 91.8%; balanced accuracy = 87.4%	Local LLM deployment is feasible for sensitive clinical text. [33]

from risk to no risk is more simpler than separating low, moderate, severe, or imminent risk. This is important for the proposed system because the system requires risk-level prediction, not only safe or unsafe classification.

The results of the SafeUT study reveal that there is a significant trade-off between precision and sensitivity. The RoBERTa neural model improved sensitivity from 39.52 to 62.02 when compared with the TF-IDF Logistic Regression baseline [26]. However, precision dropped from 75.56 to 59.52 [26]. From the perspective of a suicide risk screening task, such a trade-off might be reasonable if the model is only utilized for triage purposes. In this case, the false positive result will be checked manually, while the false negative one might have catastrophic consequences.

The 2024 CLPsych benchmark indicates that LLMs are more applicable to evidence extraction and explanation. The Team ISM system used an LLM with knowledge self-generation and output refinement, achieving recall of 93.5%, harmonic mean of 92.3%, and mean consistency of 96.1% for suicide-risk evidence extraction [32]. This is important in because healthcare systems need more than just labels of

high risk or not. It needs to provide reasons for making the classification like hopelessness, burden, intent, preparation, or plan.

Privacy preserving Llama-2 research model is also significant in terms of deployment. Fine tuned version of Llama-2 in German resulted in accuracy, sensitivity, specificity, and balanced accuracy of 87.5%, 83.0%, 91.8%, and 87.4% respectively in terms of detecting suicidality from psychiatric admission notes.

4 METHODOLOGY

This section describes the design and evaluation of a small proof-of-concept suicide risk detection engine based on Large Language Models (LLMs). The goal was not to develop an entire AI-based avatar system. Rather, the objective was to test the feasibility of developing a local LLM-based screening tool that would be able to identify suicide risk through classification of the transcripts. The system was created to be a supportive tool for triaging within a supervised environment. The system does not deliver any diagnosis,

treatment, or automated interventions. Instead, it delivers a prediction of risk with an accompanying evidence-based explanation.

The experiment was conducted through the use of the Python programming language in a Jupyter Notebook framework. For the purpose of serving the model, the use of Ollama was employed, which means that the data used did not get uploaded to any consumer-based artificial intelligence program. This was a crucial aspect of experiment since the dataset contained private mental health data.

4.1 System Architecture

Figure 1 shows the overall evaluation pipeline used in this study. The lightweight nature of the architecture was intentional because the available dataset contained only 10 examples. Fine tuning an LLM on this number of examples would not be feasible and the results would most definitely overfit the data. Also, for this PoC, we performed inference based on prompts only.

4.2 Dataset and Ethics

The PoC has used the data file provided by the lecturer [34]. The necessary data can be found on the Assignment_Cases sheet, which consists of three primary columns: Case ID, Paraphrased Dialogue, and Risk Level. The dataset includes paraphrased human-AI dialogues concerning mental health issues, classified according to five risk levels regarding suicides. Ethical considerations tell that the cases have been obtained from the Human-AI-Dialogue-Suicide-Risk-Dataset and cannot be distributed further. The dataset was analyzed locally only and not uploaded anywhere publicly.

4.3 Candidate A: Local Prompt Based LLM Classifier

Candidate A used a locally hosted Qwen2.5 LLM through Ollama. The LLM was given the full dialogue and a structured prompt containing the five risk-level definitions. The model was instructed to do the following:

- Choose one label from safe, indicator, ideation, behavior, or attempt
- Only use evidence from the conversation
- Avoid inventing facts
- Return valid JSON
- Provide a confidence level
- Provide a short evidence sentence
- Explain why the selected label was chosen

The purpose of this system is to find out if a local decoder-based LLM can conduct suicide risk prediction and explanation in a multi class scenario without further fine tuning.

4.4 Candidate B: Hybrid LLM with Safety Rule Layer

Candidate B is an extension of Candidate A. The main one is introducing an additional deterministic safety rules layer that builds upon Candidate A. The idea behind this approach is that an LLM can under classify patients in a high-risk category, particularly in cases when the language used is ambiguous or indirect. In the case of suicide risk assessment, false negatives carry more weight than false

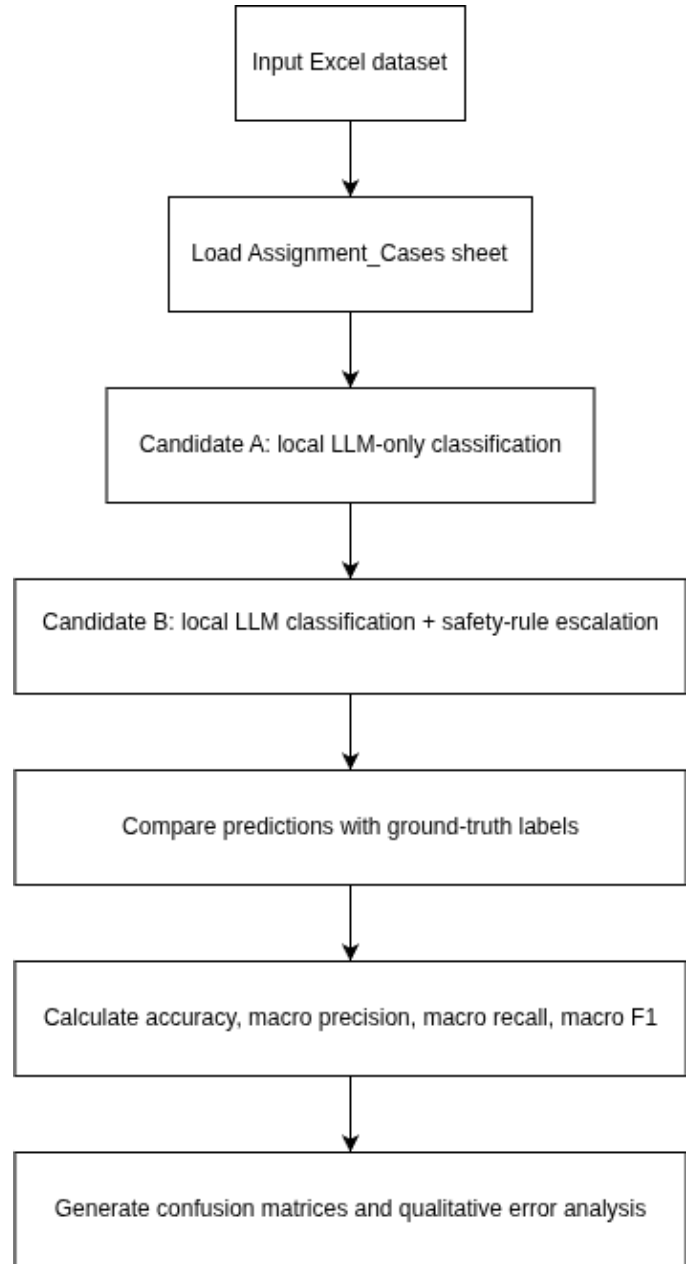


Fig. 1. Workflow of the proposed evaluation pipeline.

positives because missing someone at a higher risk is far more dangerous than wrongly classifying someone in a high-risk category. So, the hybrid model applied conservative safety rules to escalate the conversation whenever there were explicit high-risk messages detected.

4.5 Edge Deployment Environment

The proof of concept experiment was conducted using an edge computing platform using the NVIDIA Jetson AGX Orin developer kit. This type of edge computing environment is significant in the health application proposed in this project. When developing a suicide risk assessment model, there shouldn't be any communication between private individuals and public AI models. Using local or institution controlled edge infrastructure can increase the privacy aspect of such an application as no personal information gets

TABLE 3
Performance Comparison of Candidate Models

Model	Acc.	Prec.	Rec.	F1
Local LLM-only	0.50	0.54	0.50	0.49
Hybrid LLM + Safety Rules	0.80	0.87	0.80	0.79

TABLE 4
Tools and Environment Used in the Proposed System

Component	Tool
Data handling	Python, pandas
LLM inference	Ollama local endpoint
Model	llama3.2
Evaluation	scikit-learn
Visualisation	matplotlib
Environment	Jupyter Notebook / Python scripts
Deployment hardware	NVIDIA Jetson AGX Orin edge device

out from the organization. Another reason why the edge computing environment is vital is because of deployment awareness. A Jetson style device could be used for local inference in clinics, research labs, secure kiosks, or private healthcare networks.

4.6 Results

Table 3 compares the two candidate models using accuracy, macro precision, macro recall, and macro F1. Candidate A had moderate performance. It correctly classified the two safe cases and some lower or mid-risk cases. It under-classified several higher-risk cases. For example, both attempt cases were initially predicted as ideation, and one behavior case was also predicted as ideation. Local LLM was able to detect only suicidal content but struggled to find difference in ideation, behavior or attempt-level risk.

Candidate B has better performance compared to Candidate A. This is because of adding deterministic safety escalation. It correctly classified explicit high risk cases that Candidate A could not classify. For example, cases where the LLM predicted ideation were escalated to attempt or behavior when the rule layer detected stronger risk signals. This improved macro F1 from 0.49 to 0.79. However, there are some disadvantages as well. The system classified one indicator case as ideation because the text contained direct death related language even though the truth label was lower. This clearly shows how safety rules contribute to the problem of over-escalation.

4.7 Implementation and Reproducibility

Table 4 summarizes the main tools, model, software environment, and deployment hardware used in the proposed system.

5 AI ETHICS

The system aligns with human and social well being principle because it aims to identify suicide risk signals earlier and refer high risk users to mental health professionals [35]. Environmental impact principle is also considered because this system used a local edge device and a compact local

LLM. There is no utilization of large cloud based system. In production system, model size, energy use should be monitored carefully to reduce unnecessary carbon impact. The system also respects user dignity, autonomy and safety. Users are informed that AI system is being utilized and this does not provide a clinical diagnosis. This system will not pretend to be a human therapist or make false claims. Its role is limited. It only identifies possible risk signals and gives referrals to human professionals. Fairness is a major concern here because suicide risk may be expressed differently by people according to age, gender, culture, language, disability and communication style. The given dataset is small to prove everything. Future validation is necessary to remove false positives and false negatives across various demographic groups.

This project handles privacy well. The system data will contain sensitive mental-health content and any leak to these data will have serious consequences. For this reason the system is based on local inference. Ollama is being used locally on an edge device of Nvidia Jetson Orion. The system is also able to run without any internet connections. There is no expose to public API or servers. For reliability and safety the proof of concept showed that the local LLM-only model under classified some higher risk cases. However, hybrid LLM improved its performance. This shows that only prompt based LLM is not safe enough for suicide risk detection. The system also escalates uncertain classification to a qualified professional for human review. The system improves transparency by returning a risk label, confidence level, supporting evidence, and explanation. The hybrid model also reports whether a safety rule was triggered. This allows health professionals to understand why a case was escalated rather than relying on a black-box label. For contestability, health professionals will be able to override AI classifications, and users will be able to request human review where necessary. The AI model itself cannot be accountable for clinical outcomes. Therefore, responsibility will remain with the organization deploying the system and the qualified professionals supervising it.

6 DISCUSSION

This system aligns really well with UN Sustainable Development Goal 3: Good Health and Well-being [36]. Goal 3 aims to ensure healthy lives and promote well-being for all at all ages [36]. This goal also includes promoting mental health and well-being [36]. The proposed system supports this goal by exploring how AI can assist in early detection of suicide risk. It help to support people with mental illness by faster referral to qualified healthcare professionals. This system also supports UN Sustainable Development Goal 9: Industry, Innovation and Infrastructure [36]. This is because investigates a practical AI architecture for privacy-preserving mental-health screening. The use of local LLM pipeline on an edge device is a novel concept. It also shows how AI can be innovated on sensitive healthcare environments without relying on public cloud APIs. This is extremely relevant because AI in healthcare needs trustworthy digital infrastructure, secure deployment, and responsible integration. This system also has some relevance to Sustainable Development Goal 9: Industry, Innovation and Infrastructure [36]. This is

because accessible screening tools may help identify at-risk individuals earlier, particularly where mental-health services are overloaded or difficult to access.

The carbon footprint of this system is very low compared to cloud based LLM API providers. This is because The PoC was run locally on a Jetson AGX Orin edge device using Ollama. The edge device consumes lower power compared to normal electronic devices. Inference was done locally and privately. Specifically, the Ollama model was deployed locally on the edge device, and private conversations were not uploaded to publicly available consumer AI applications like ChatGPT. This was intentional since transmitting private suicide-risk conversations to the API would be a privacy and legal issue.

7 CONCLUSION

This report investigated how AI models specifically Large Language Models can be used for suicide risk detection from conversational data in a clinically responsible way. The problem was to investigate whether AI models were able to support early screening and referral while managing the safety, privacy, ethical, and operational risks. The report explored various AI system from keyword based to large language models. The benchmarking experiments conducted on similar data sets revealed that suicide risk classification is challenging and accuracy on high-risk cases is more important than overall accuracy. In proof of based system, hybrid system was found to be more better and accurate than only plain prompt based local LLM. A key implication is that LLMs should only serve in a role as decision support systems, rather than automated suicide risk assessors. Further research should consider a much wider range of clinically validated datasets, investigate other local large language models and encoders, assess demographic fairness, enhance calibration, and validate the framework with clinicians prior to practical application.

APPENDIX 1 - VIDEO PRESENTATION

<https://echo360.net.au/media/0b9266e9-aa14-41bc-b5c1-556157017cde/public>

APPENDIX 2 – SUPPLEMENTARY MATERIALS

https://drive.google.com/drive/folders/12KQxhQf2hXi0VoPEizzD61IY_uQ-pU5n?usp=sharing

https://github.com/PrasannaTandukar/COMP6011_AS2

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